



United International University (UIU)
Department of Computer Science and Engineering (CSE)
Course: VLSI Design (CSE 4327)

Experiment 2: Layout Design of Standard Cell (Inverter)

Objectives

1. To understand the concept of a standard cell layout and how it differs from a transistor-level schematic.
2. To become familiar with the Cadence Virtuoso Layout XL editor and its core drawing, alignment, and routing tools.
3. To translate a verified CMOS inverter schematic into a physical layout that follows fixed-height standard-cell architecture rules.
4. To verify the physical design using PVS (Physical Verification System) - running LVS (Layout vs. Schematic) and DRC (Design Rule Check) - and to correct any reported violations.

Introduction

A standard cell layout is the physical representation of a logic gate, transforming the abstract transistor-level schematic into a set of geometric patterns. While the schematic focuses on functional relationships, the layout defines the physical characteristics of the circuit - such as the size, position, and interconnects of layers - that will be fabricated on a silicon wafer. In a standard cell methodology, the layout must adhere to strict “cell architecture” rules, such as a fixed height and standardized power rail positions, to ensure that multiple cells can be placed and routed automatically by EDA tools. For this lab, we use Layout XL and PVS (Physical Verification System) within the Cadence Virtuoso suite to translate our verified inverter schematic into a physical design.

Note: This experiment continues directly from Experiment 1 of the schematic design lab. Make sure your CMOS inverter schematic has been checked and saved with no errors before proceeding, since Layout XL will use it as the source for connectivity.

Part 1: Opening Layout XL and Creating the Layout Cellview

1. Once the schematic is verified, proceed to Layout XL. From the schematic window, go to Launch → Layout XL.

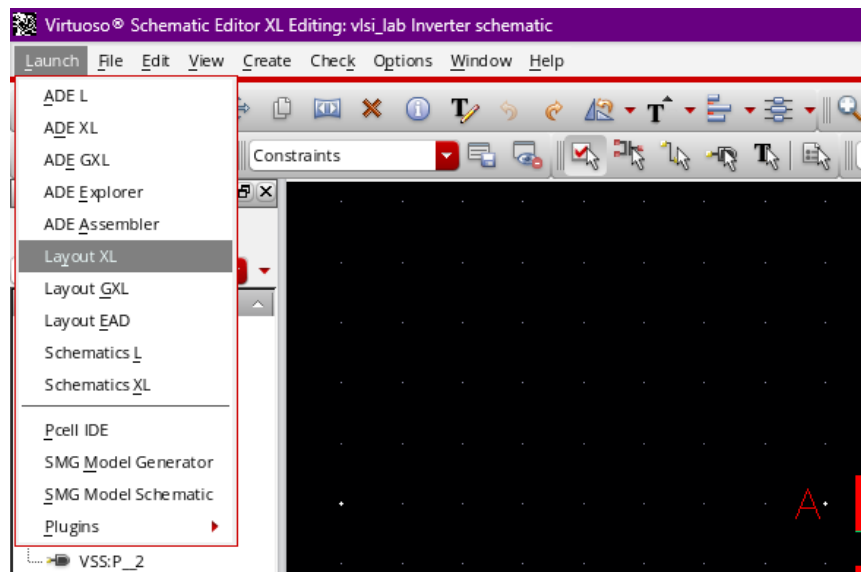


Figure 1: Launch menu in the Schematic Editor, with Layout XL selected.

2. In the New File form, make sure Library is set to your design library, the Cell is set to inverter, and View and Type are set to layout. Confirm the Library path file points to the cds.lib in your work area and click OK.

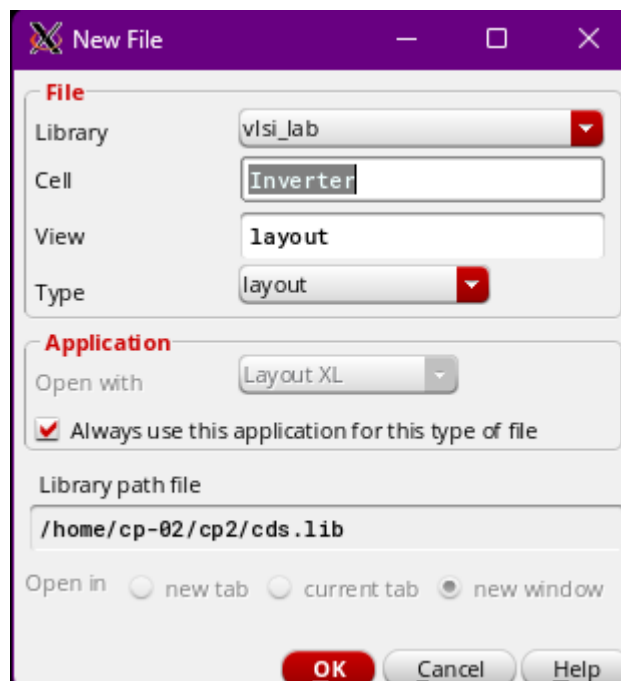


Figure 2: New File form for creating the inverter layout cellview.

3. A blank Layout Suite XL window opens for the inverter layout, with the technology layer palette listed on the left.

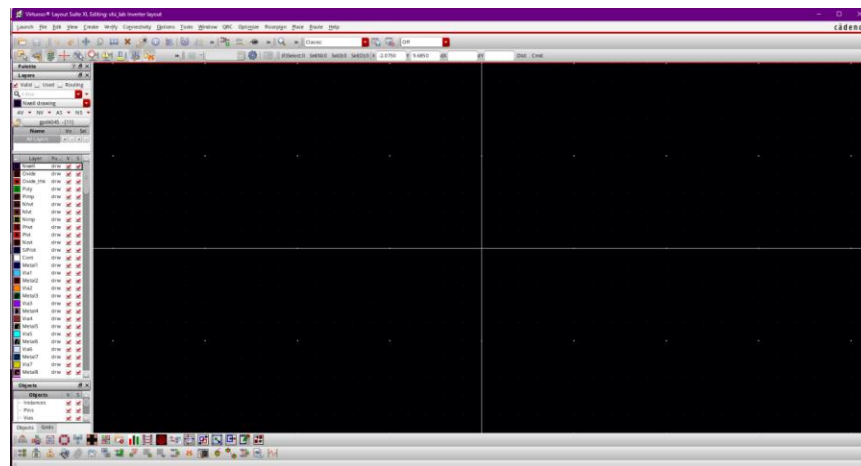


Figure 3: Blank Layout Suite XL window with the gpd045 layer palette loaded.

Part 2: Generating the Layout from the Schematic

4. Use Connectivity → Generate → All From Source. This reads the connectivity of the verified schematic and prepares the corresponding devices and pins for the layout.

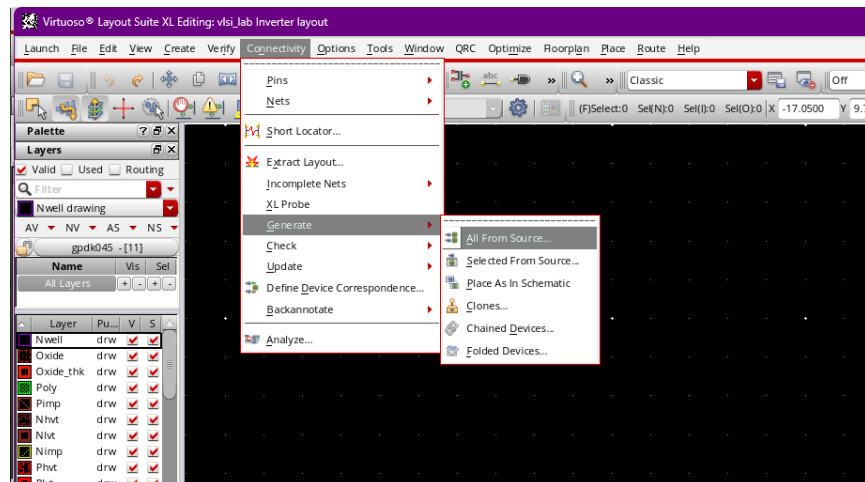


Figure 4: Connectivity → Generate → All From Source menu.

5. In the “Generate Layout” form, switch to the I/O Pins tab. Change the Layer from Metal1 drawing to Metal1 pin, then click Apply. Set both the Width and Height to 0.06, and ensure the Create Label As checkbox is selected and set to Label.

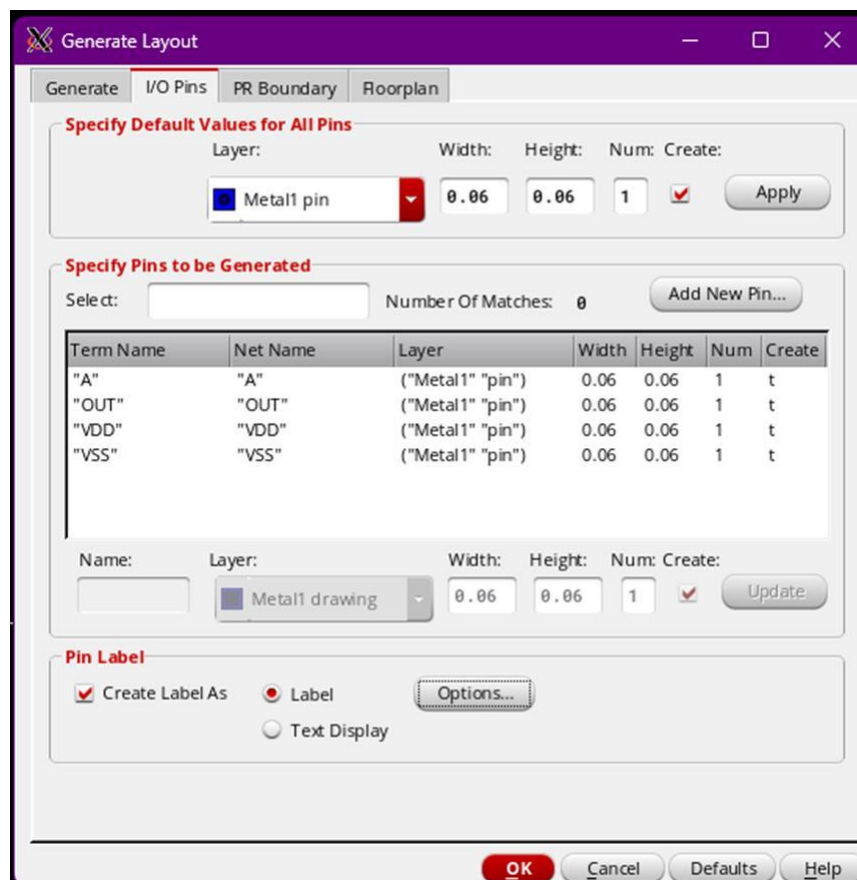


Figure 5: Generate Layout form, I/O Pins tab, with Metal1 pin selected and the A, OUT, VDD and VSS pins listed.

6. Select Options and set the Pin Label Height to 0.05, set Layer Name to 'Same As Pin' and Layer Purpose to 'label', then click OK to generate the layout components and pins in the layout editor.

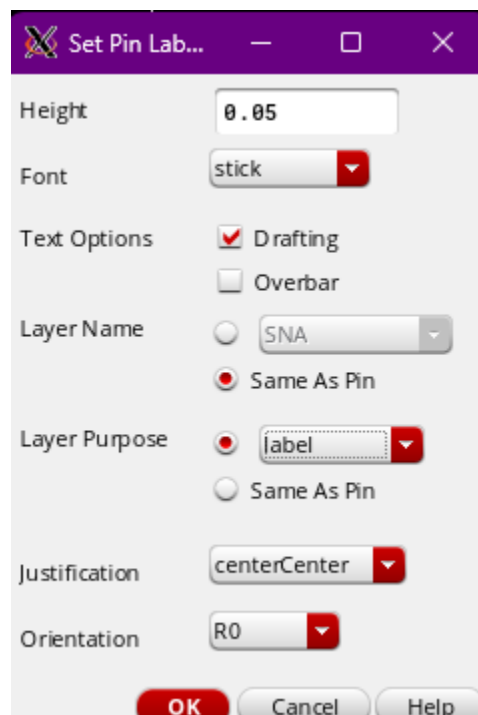


Figure 6: Set Pin Label form.

7. The generated layout will initially appear as colored component outlines or boxes, representing the pmos1v and nmos1v devices at an abstract level.

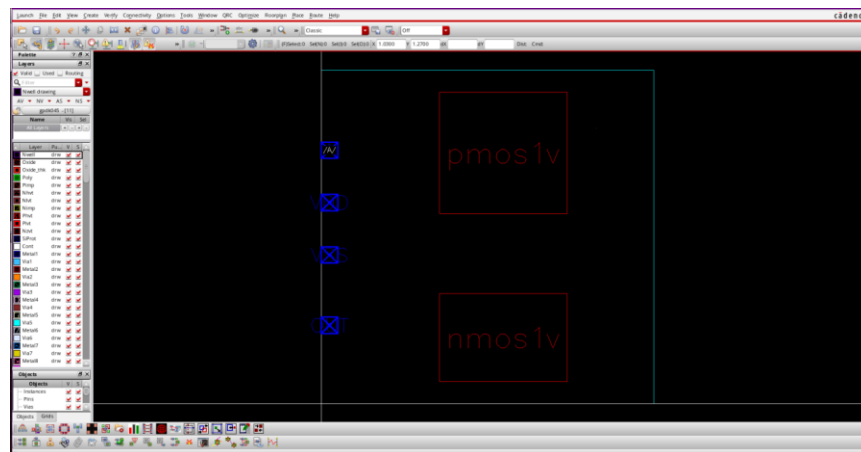


Figure 7: Generated layout shown as abstract component outlines (pmos1v and nmos1v).

8. Press Shift+F on your keyboard. This toggles the display from abstract boxes to the detailed physical view, revealing the internal layers (active regions, polysilicon, contacts).

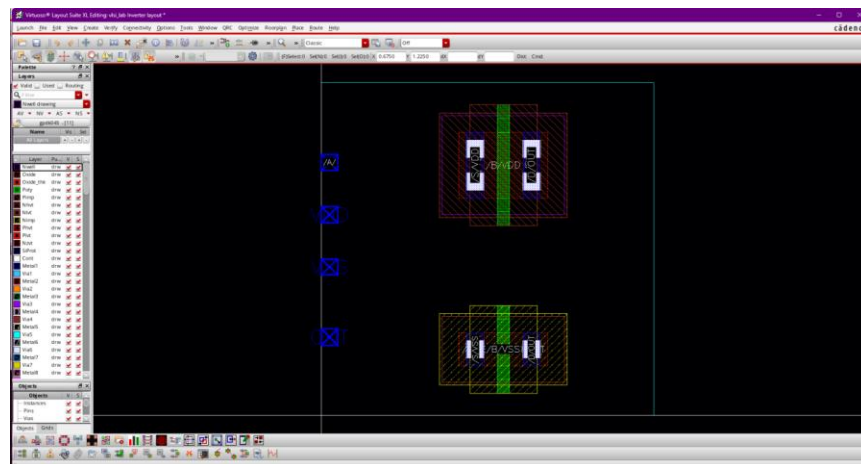


Figure 8: Detailed physical view after pressing Shift+F, showing active, poly and contact layers.

9. From the toolbar, ensure that Partial Select mode is enabled and Orthogonal Mode is turned ON. These settings make precise selection and manual routing much easier.

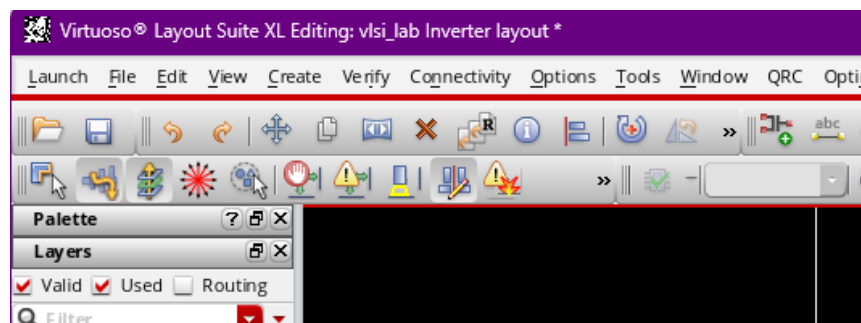


Figure 9: Toolbar with Partial Select and Orthogonal Mode controls highlighted.

Part 3: Defining the Standard-Cell Height and Floorplan

10. Press K to enable the ruler tool. Click at the bottom origin and measure a fixed vertical standard cell height (e.g., 1.8 μm).

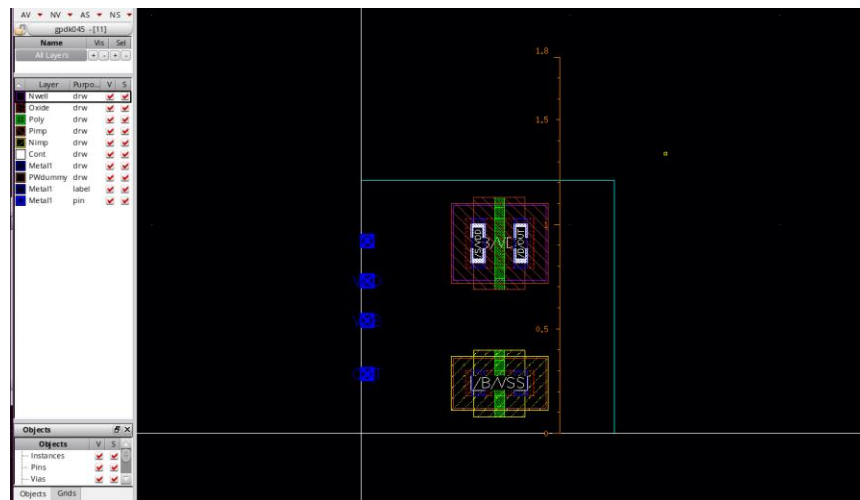


Figure 10: Ruler tool measuring a 1.8 μm standard-cell height from the origin.

11. Now press A to activate the Align tool. Click on the upper boundary of the PR (Placement and Routing) boundary, then click on the 1.8 μm ruler reference line to snap the PR boundary to the target cell height.

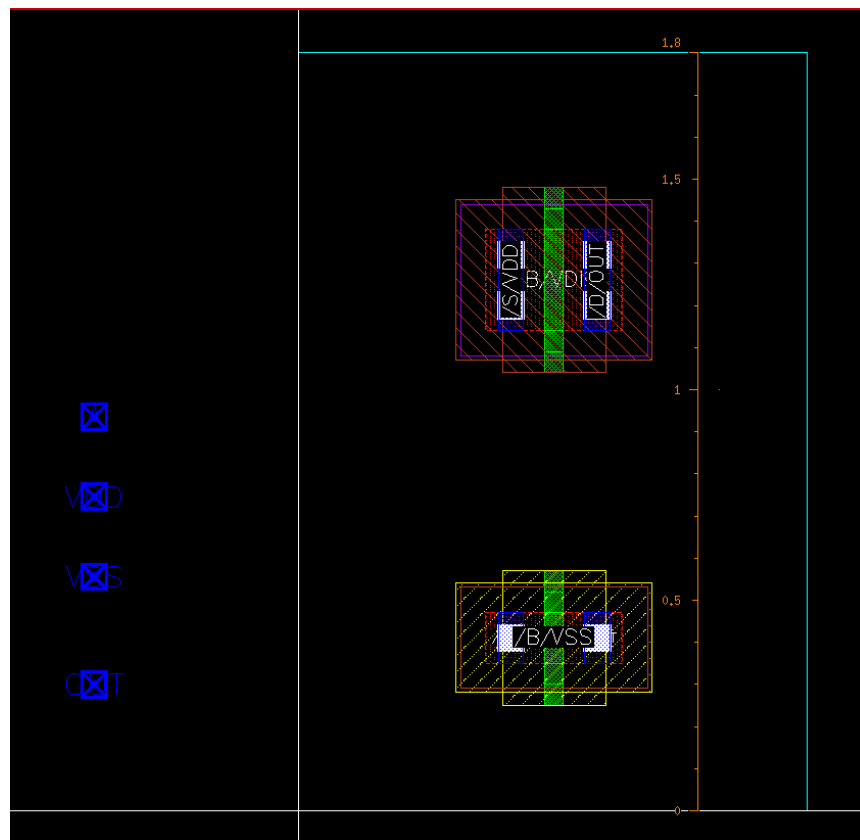


Figure 11: PR boundary aligned to the 1.8 μm ruler reference line.

12. Select the Metal1 layer from the layer palette.

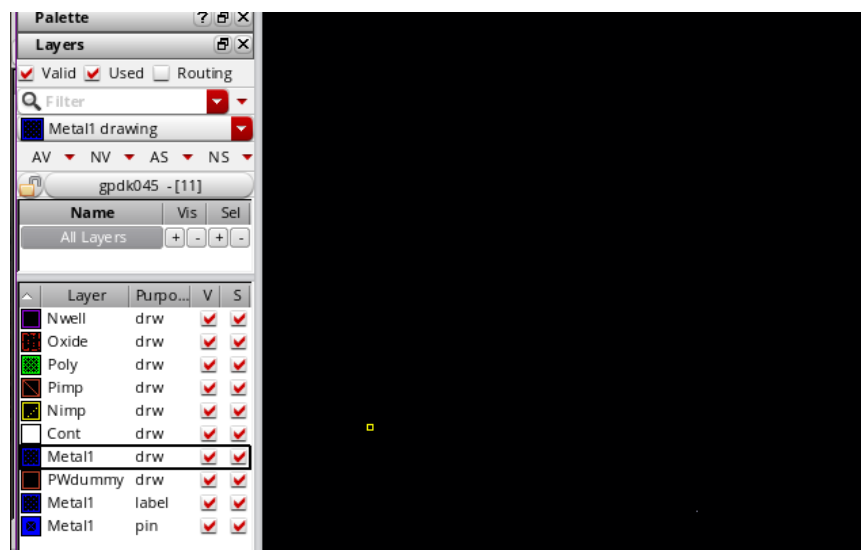


Figure 12: Metal1 drawing layer selected in the layer palette.

- Press P to draw a path for the VDD rail (top) and VSS rail (bottom) along the top and bottom edges of the PR boundary.



Figure 13: VDD and VSS power rails drawn as Metal1 paths along the PR boundary.

- Then select both VDD and VSS rails. Press Q to change the Width of both rails from 0.06 μm to 0.12 μm . Align the top edge of the VDD rail and the bottom edge of the VSS rail with the top and bottom limits of your PR boundary.

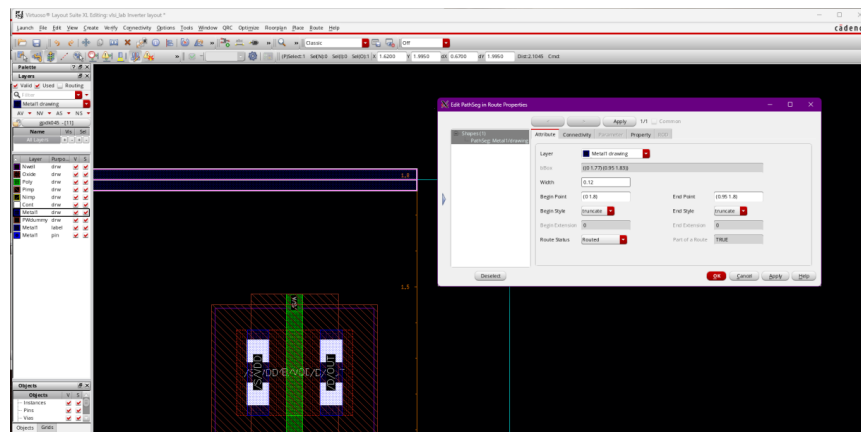


Figure 14: Edit PathSeg in Route Properties form, updating the rail width to $0.12\ \mu\text{m}$.

15. Press K to activate the ruler tool. Measure $0.1\ \mu\text{m}$ down from the bottom edge of the top VDD rail. Move and align the top edge of your PMOS transistor active region to this $0.1\ \mu\text{m}$ mark.
16. Press K again. Measure $0.1\ \mu\text{m}$ up from the top edge of the bottom VSS rail. Move and align the bottom edge of your NMOS transistor active region to this $0.1\ \mu\text{m}$ mark.

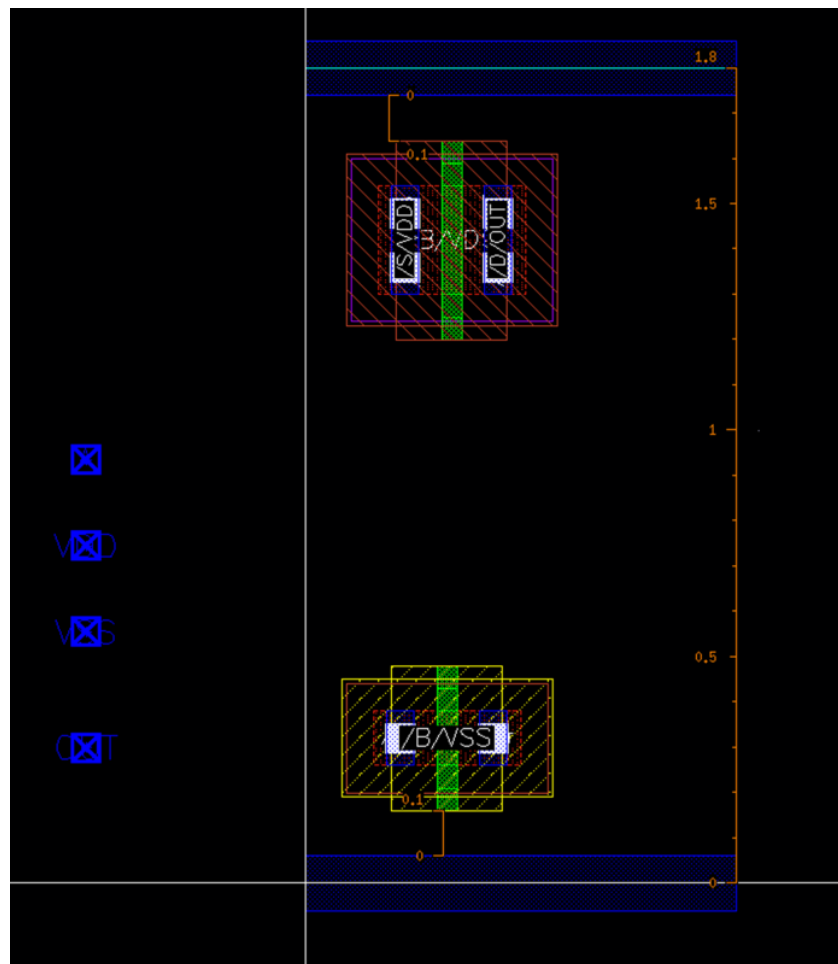


Figure 15: PMOS and NMOS active regions aligned $0.1\ \mu\text{m}$ in from the VDD and VSS rails.

Part 4: Aligning the Transistors and Detaching the Body Taps

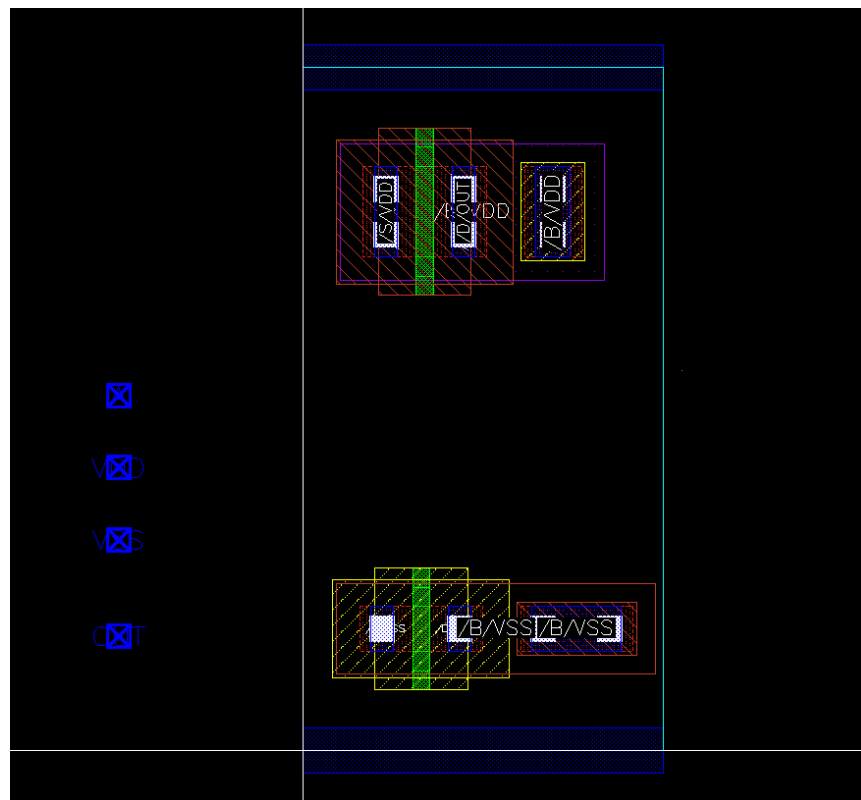


Figure 18: Layout with rulers cleared, showing the aligned PMOS and NMOS gates with detached body taps.

Part 5: Adding the N-Well and Implant Layers

21. Press K to draw a ruler extending $0.2\ \mu\text{m}$ out to the left of the PMOS and NMOS active regions.
22. Use the ruler tool to measure down from the top edge of the PR boundary to a vertical line at the $0.9\ \mu\text{m}$ mark (the exact center point of the overall $1.8\ \mu\text{m}$ cell height).
23. Use the ruler tool from the bottom edge to confirm the remaining $0.9\ \mu\text{m}$ area.

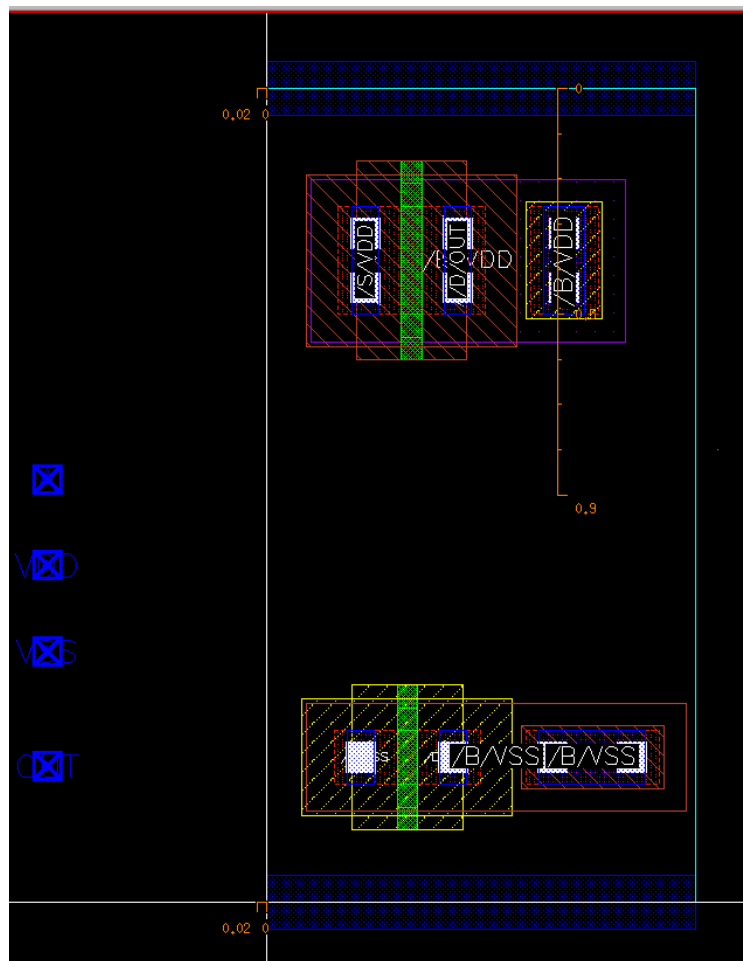


Figure 19: Reference rulers at $0.2\text{ }\mu\text{m}$ from the transistors and at the $0.9\text{ }\mu\text{m}$ cell mid-line.

24. Select the NWELL layer from the Palette and press R to activate the Rectangle tool.

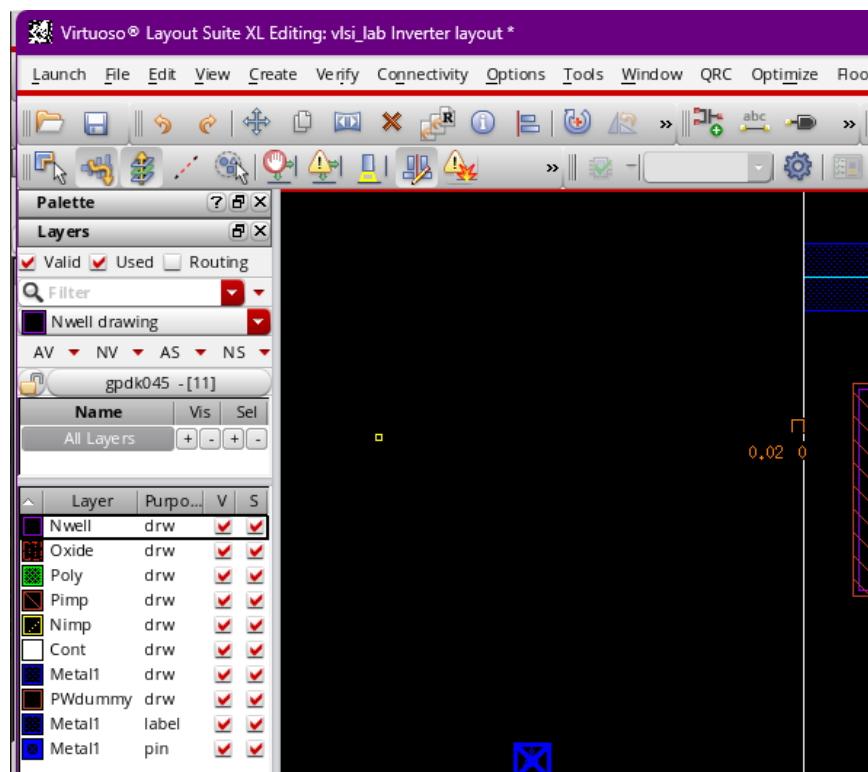


Figure 20: Nwell drawing layer selected, with the Rectangle tool active.

25. Draw an N-Well rectangle starting from the 0.2 μm outer left margin of the PMOS, extending up to the top PR boundary and down to the central 0.9 μm dividing line.

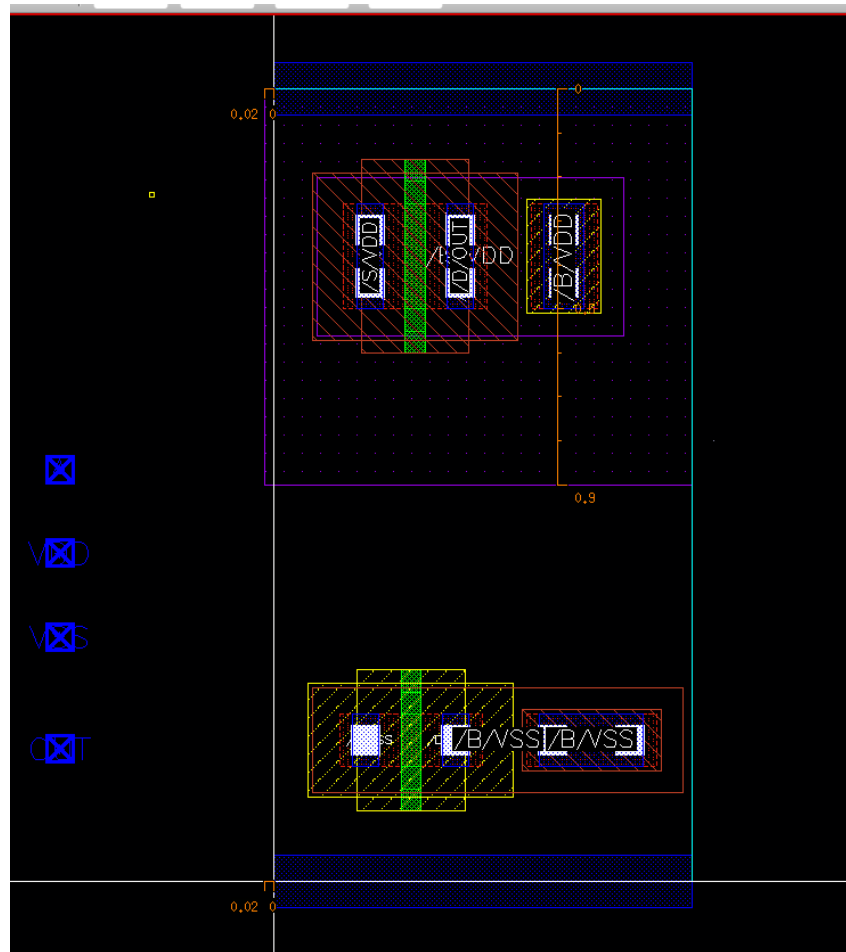


Figure 21: N-Well rectangle drawn over the PMOS (upper) half of the cell.

26. Select the Nimp layer from the Palette.

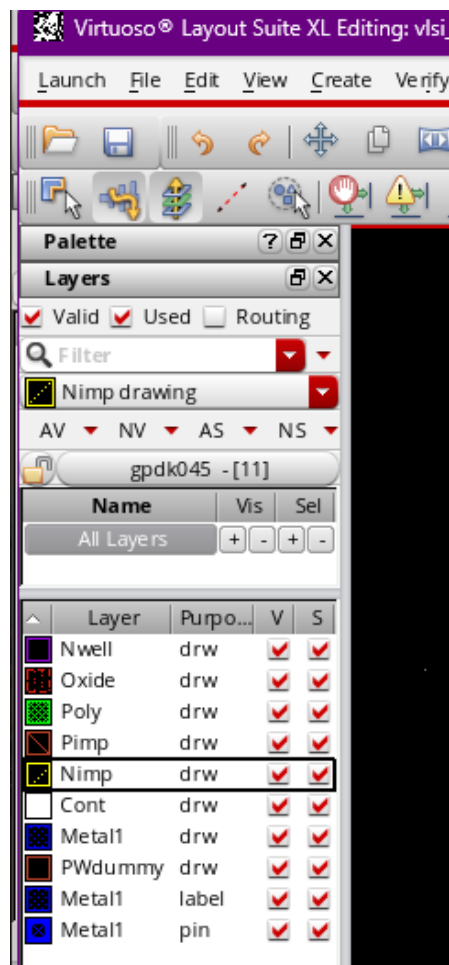


Figure 22: Nimp drawing layer selected in the layer palette.

27. Press R to draw a rectangle over the NMOS region, extending from the 0.2 μm outer left margin down to the bottom and up to the central 0.9 μm dividing line.

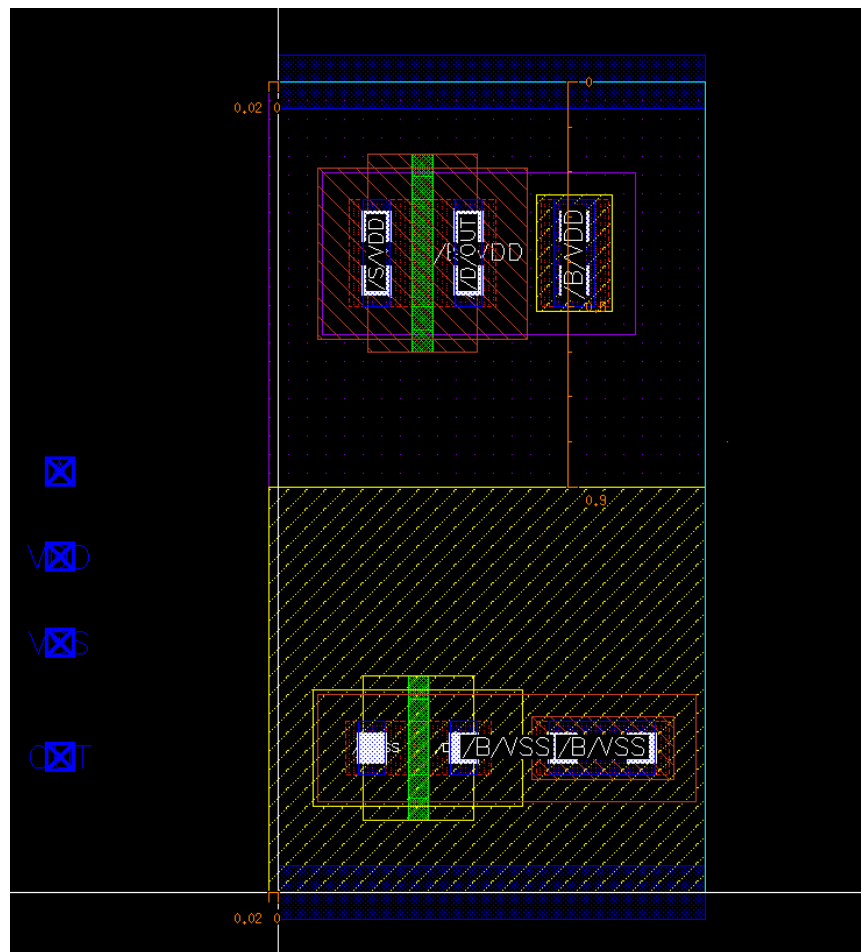


Figure 23: Nimp rectangle drawn over the NMOS (lower) half of the cell.

28. Press Shift + K to clear and erase all rulers on your layout window once the rectangles are drawn.

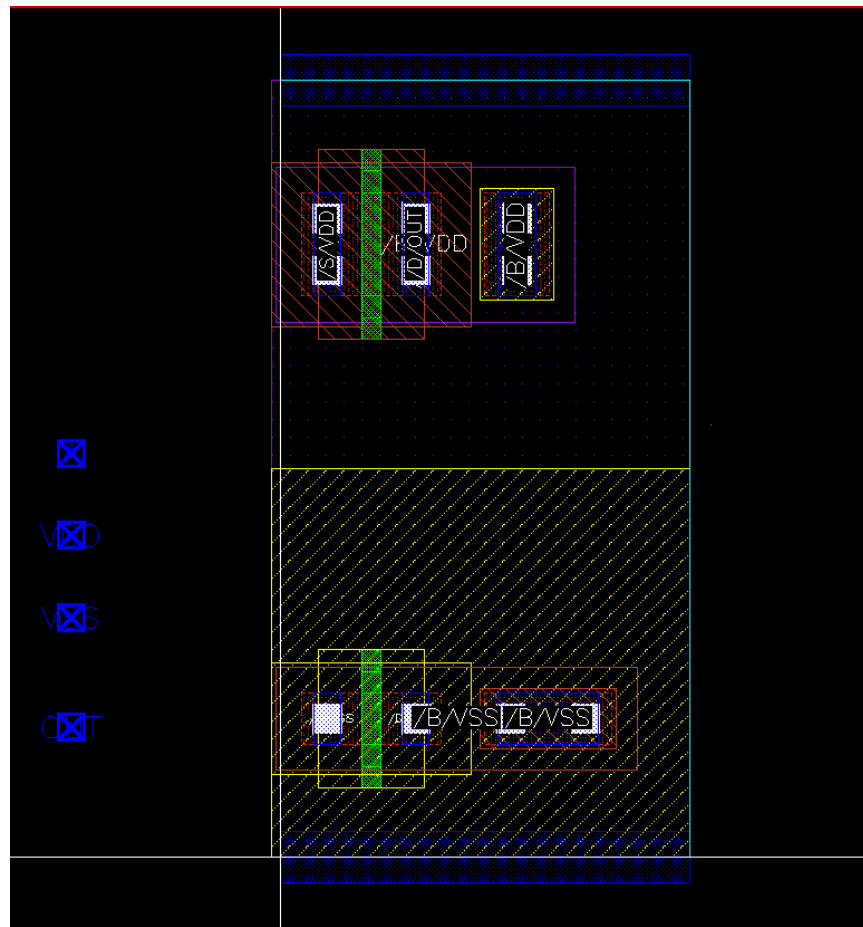


Figure 24: Layout with the N-Well and Nimp regions drawn and all rulers cleared.

Part 6: Routing the Gate, Supply and Output Nets

29. Next, select the Poly layer.

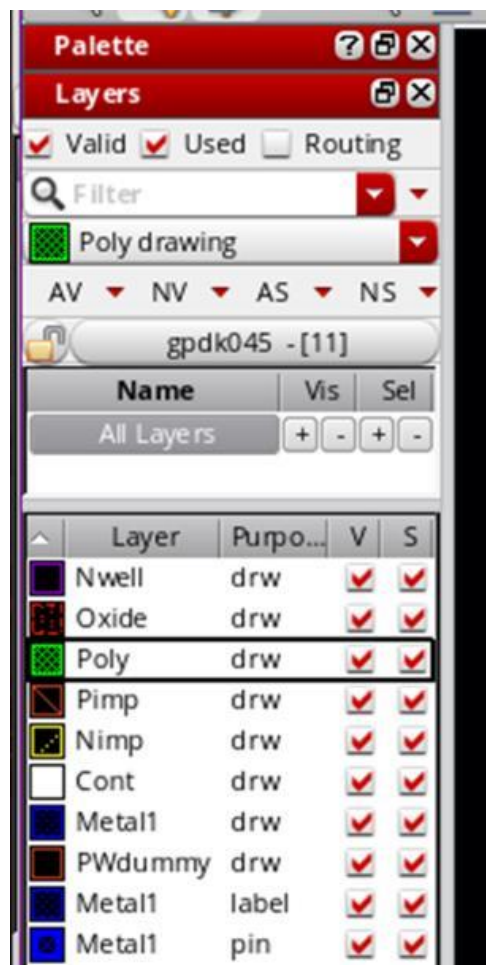


Figure 25: Poly drawing layer selected in the layer palette.

30. Press P to draw a routing path that directly shorts the PMOS and NMOS gates together vertically, forming the common input node.

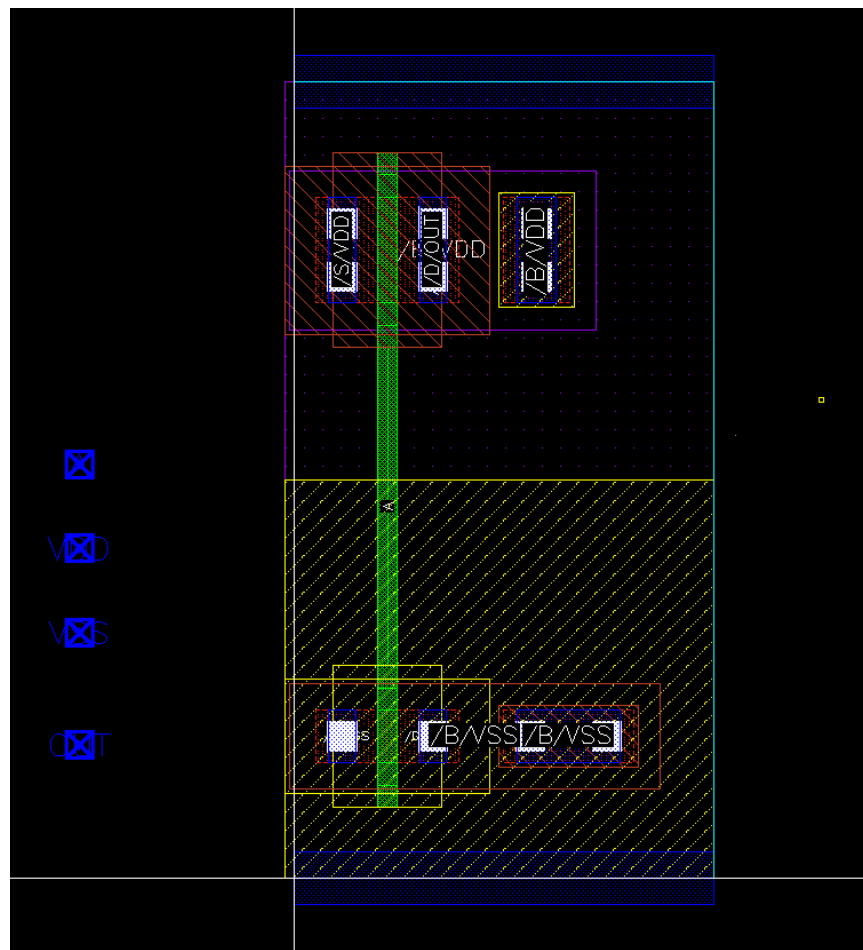


Figure 26: Vertical Poly routing path shorting the PMOS and NMOS gates.

31. Select the Metal1 layer.

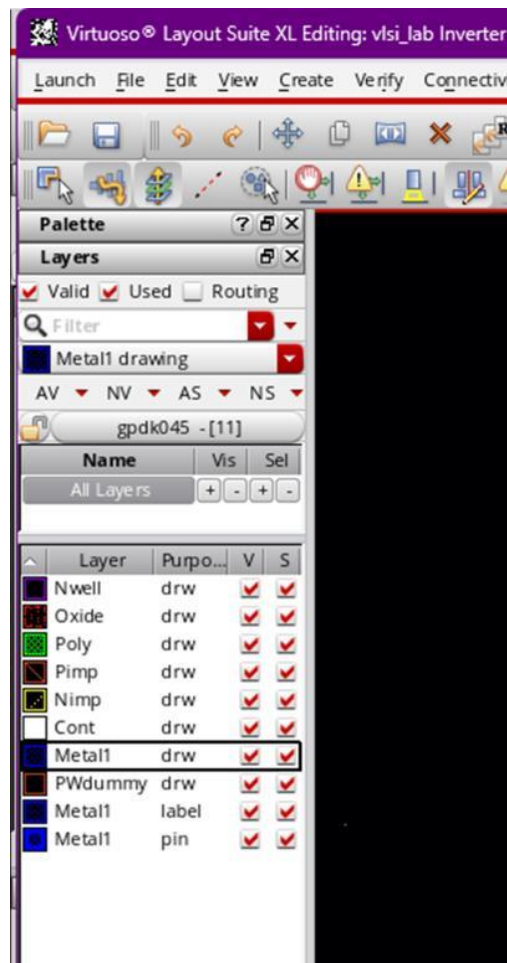


Figure 27: Metal1 drawing layer selected in the layer palette.

32. Press P to connect the PMOS source to the VDD rail, and route a separate tie line from the VDD rail to the detached PMOS body. Route another Metal1 line to connect the NMOS source to the VSS rail, and tie the VSS rail to the detached NMOS body.

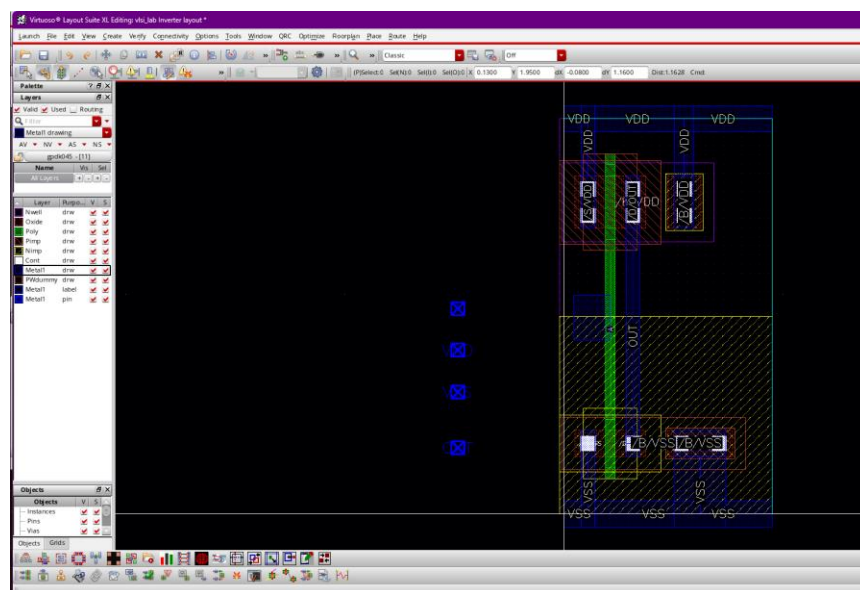


Figure 28: Metal1 tie lines connecting the PMOS/NMOS sources and detached bodies to the VDD and VSS rails.

33. Short the PMOS drain and NMOS drain together using a vertical Metal1 line to form the OUT net.

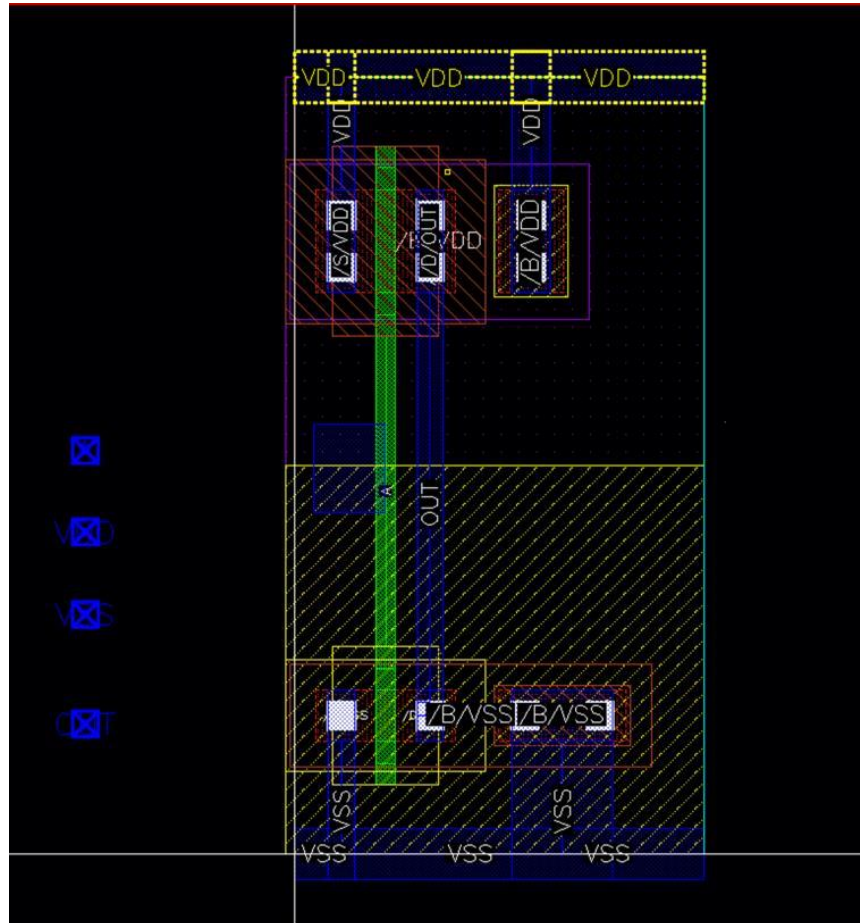


Figure 29: PMOS and NMOS drains shorted with a vertical Metal1 line, forming the OUT net.

34. Next, place VDD and VSS Metal1 pins directly onto the respective power rails.

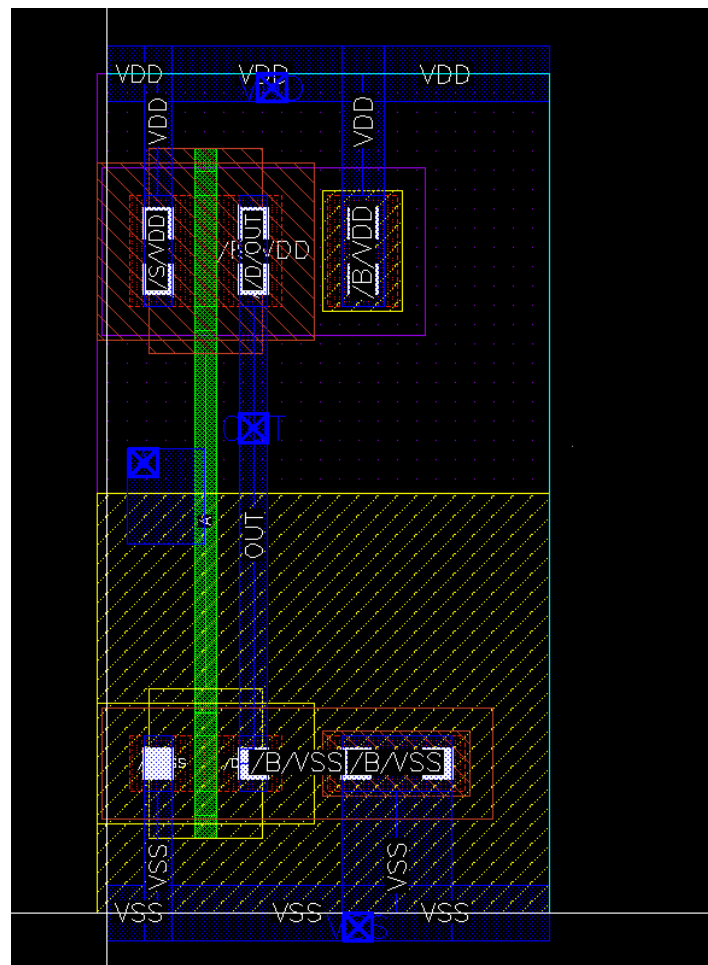


Figure 30: VDD and VSS Metal1 pins placed directly on the power rails.

Part 7: Connecting Poly to Metal with a Via

35. To connect the shared Poly gate to the Metal1 input pin, a via is required between the two layers.

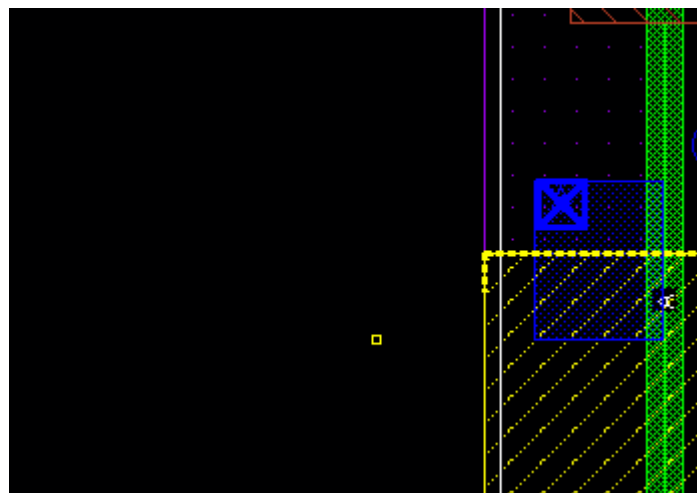


Figure 31: Close-up of the area where the Poly gate line meets the Metal1 input pin.

36. Use a via. Press O to open the Create Via form.



Figure 32: Create Via form.

37. Set the Via Definition to M1_PO (Metal1 to Poly), set Columns to 2 to form a 1 x 2 via array, then click on Hide.

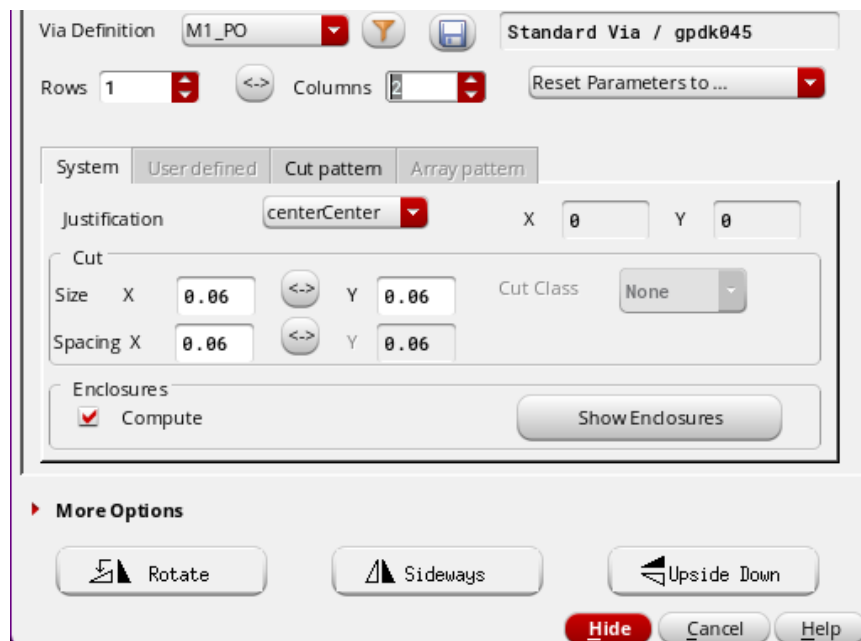


Figure 33: Via Definition set to M1_PO with Columns = 2.

38. Place this via directly onto the shared vertical Poly gate line.

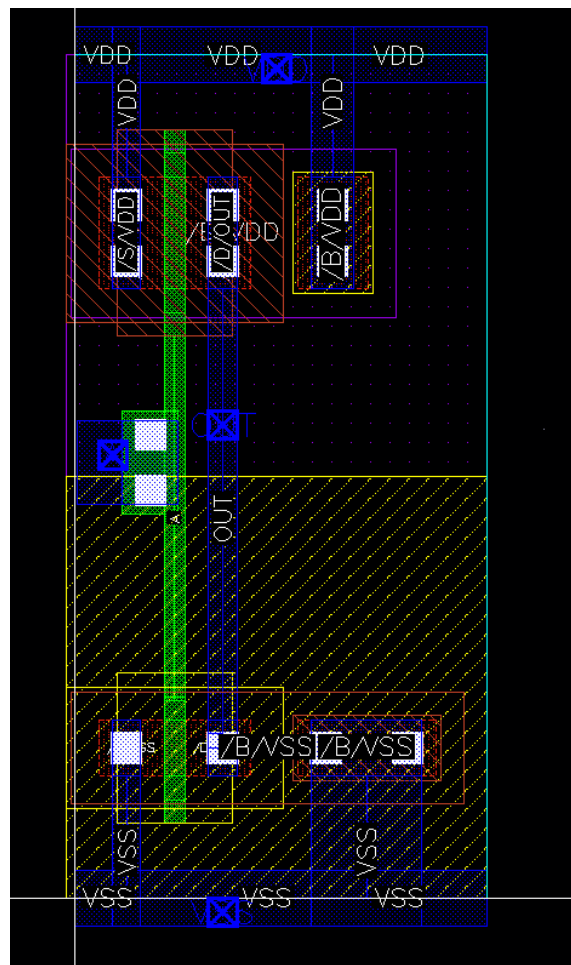


Figure 34: 1 x 2 M1_PO via array placed on the shared Poly gate line.

39. Take the Metal1 pin A and place it over the via on the Poly gate, then press R to draw a small Metal1 rectangle over the via to make it look like the rest of the layout.

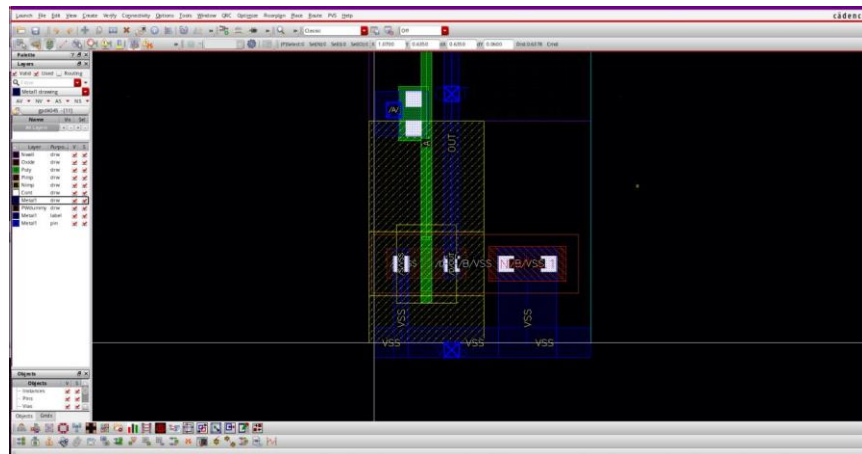


Figure 35: Metal1 pin A placed over the via, with a small Metal1 rectangle drawn to match the surrounding layout.

Part 8: Physical Verification - Running LVS

40. Go to Plugins → PVS, then choose Run LVS.

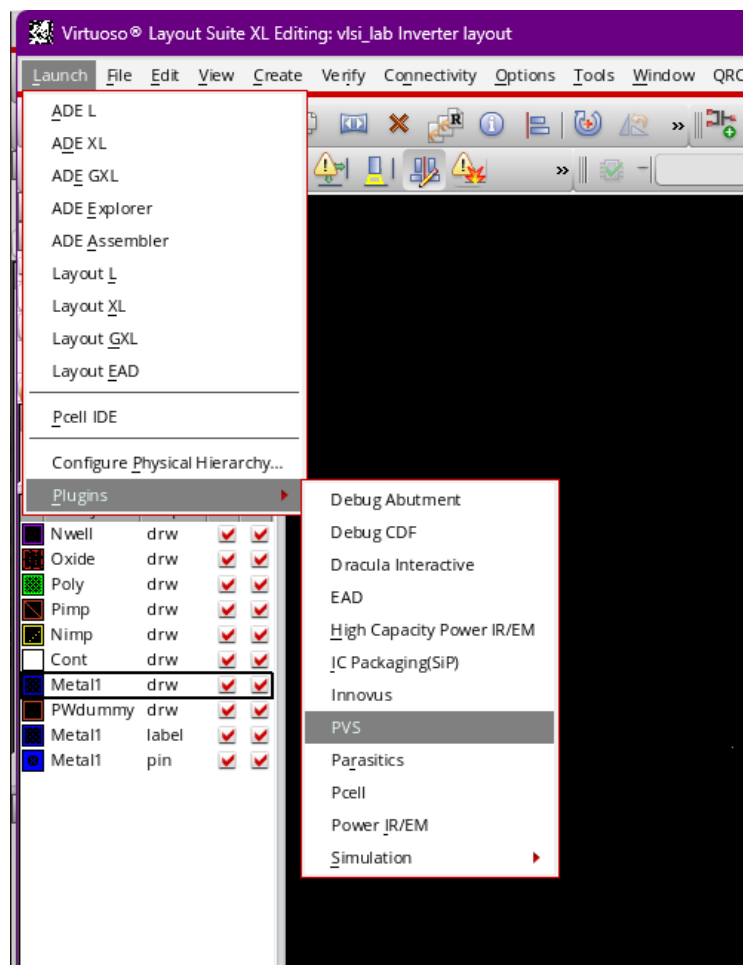


Figure 36: Launch → Plugins → PVS menu.

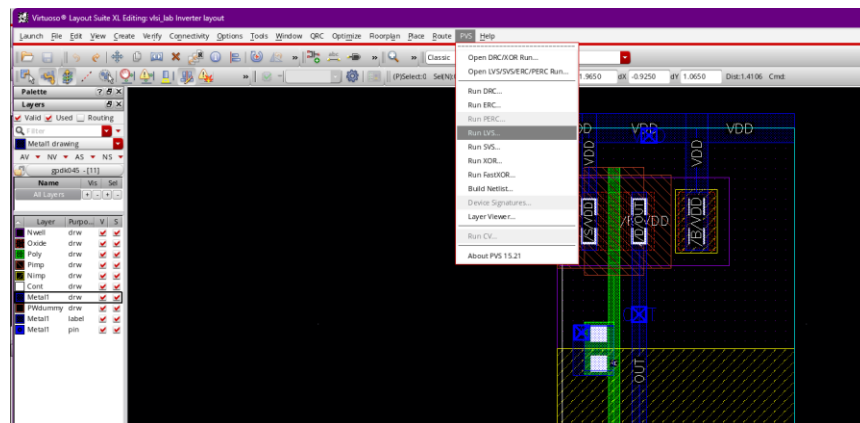


Figure 37: PVS menu with Run LVS selected.

41. In the PVS LVS Run Submission form, set the Run Directory name to inverter.

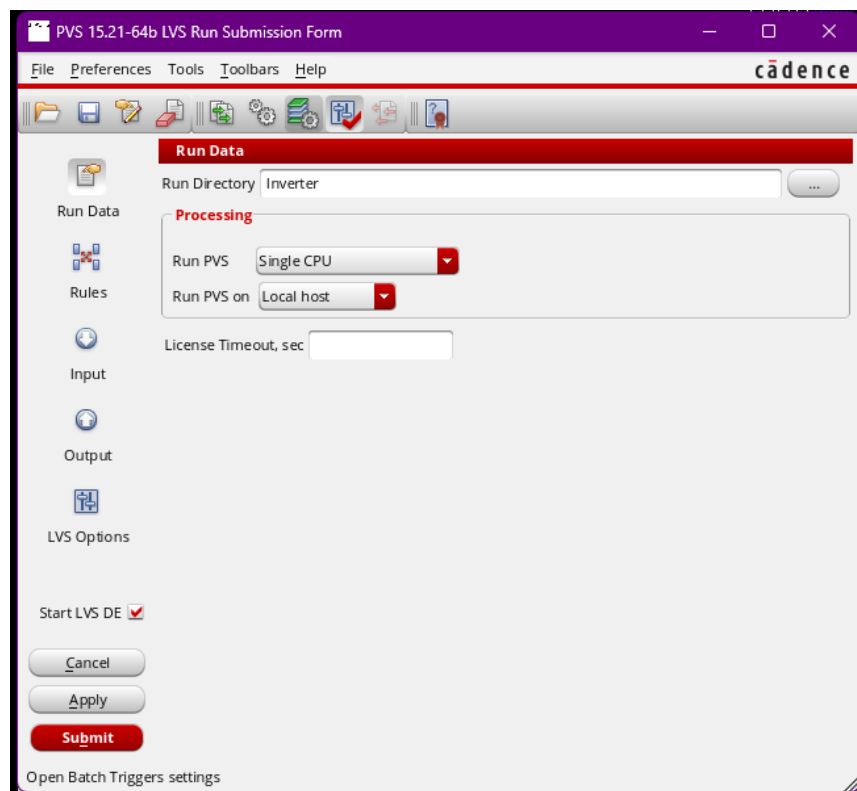


Figure 38: PVS LVS Run Submission form with the Run Directory set to inverter.

42. Switch to the Rules tab to configure your verification files. Click the three-dot button next to the Technology field, then browse to and select: /pkg/eee/cadence/tech/gpdk045_v_5_0/pvtech.lib.
43. Select gpdk045_pvs from the Technology list. Click Apply or Submit to process the LVS run.

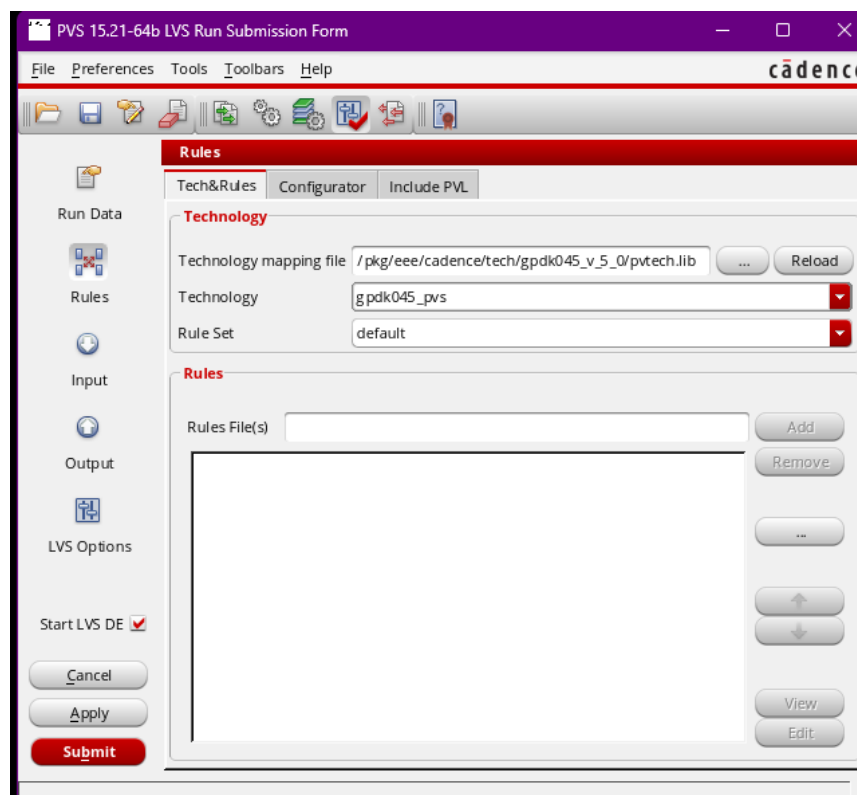


Figure 39: Rules tab with the technology mapping file and gpdk045_pvs technology selected.

44. Verify that the extractor displays a successful match (Comparison Results: Match).

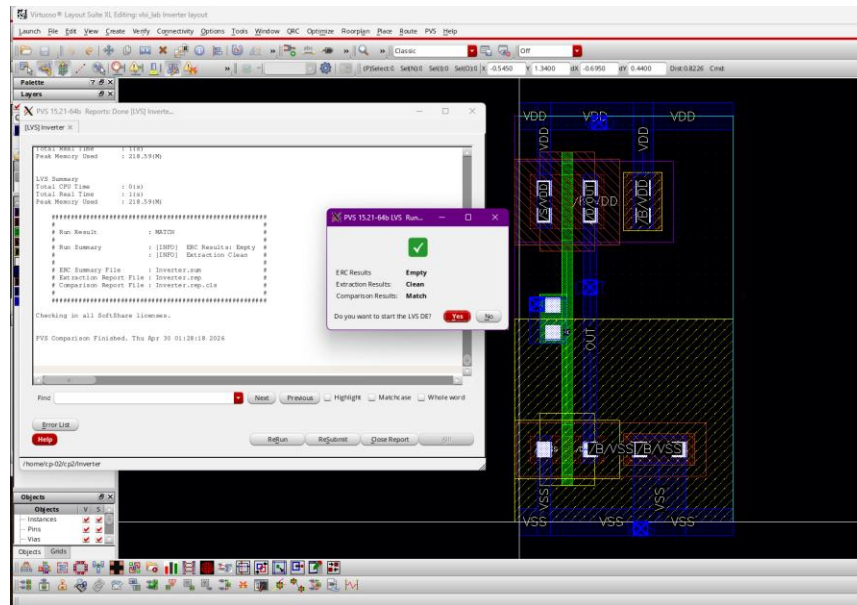


Figure 40: LVS run result showing ERC Results: Empty, Extraction Results: Clean, and Comparison Results: Match.

Part 9: Physical Verification - Running DRC and Fixing Violations

45. Go to Plugins → PVS → Run DRC.

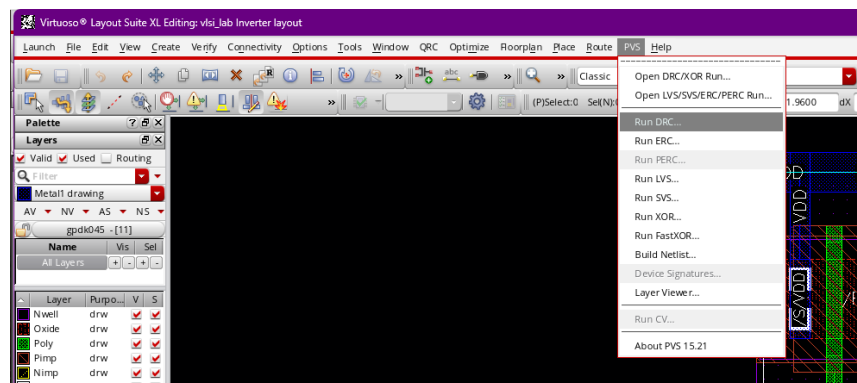


Figure 41: PVS menu with Run DRC selected.

46. Use the exact same technology path settings as for LVS (/pkg/eee/cadence/tech/gpdk045_v_5_0/pvtech.lib) and select Technology gpdk045_pvs.

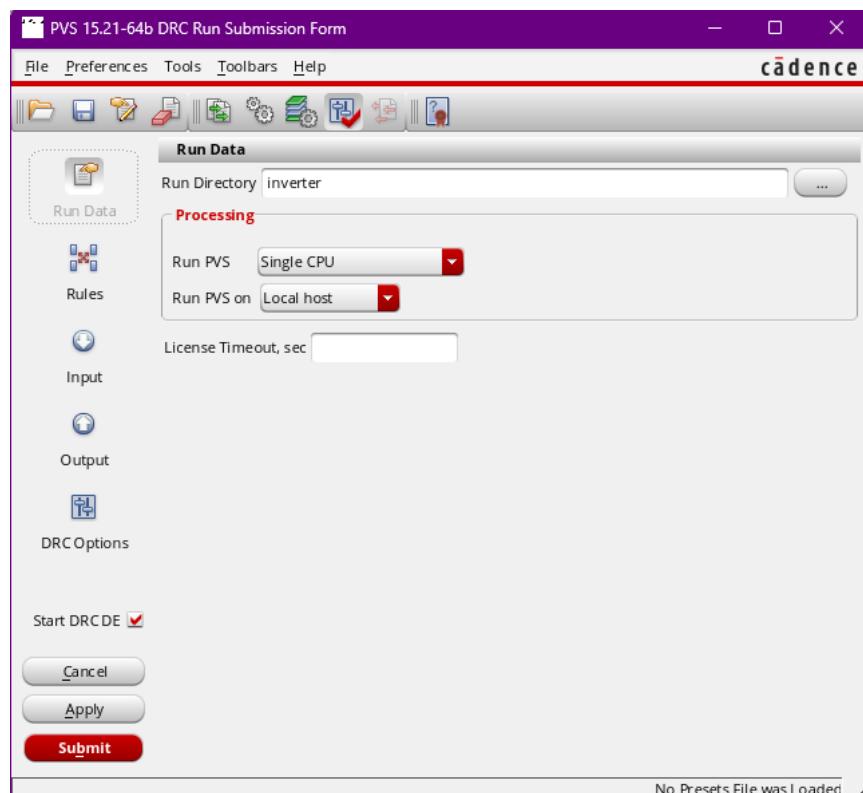


Figure 42: DRC Run Submission form with the Run Directory set to inverter.

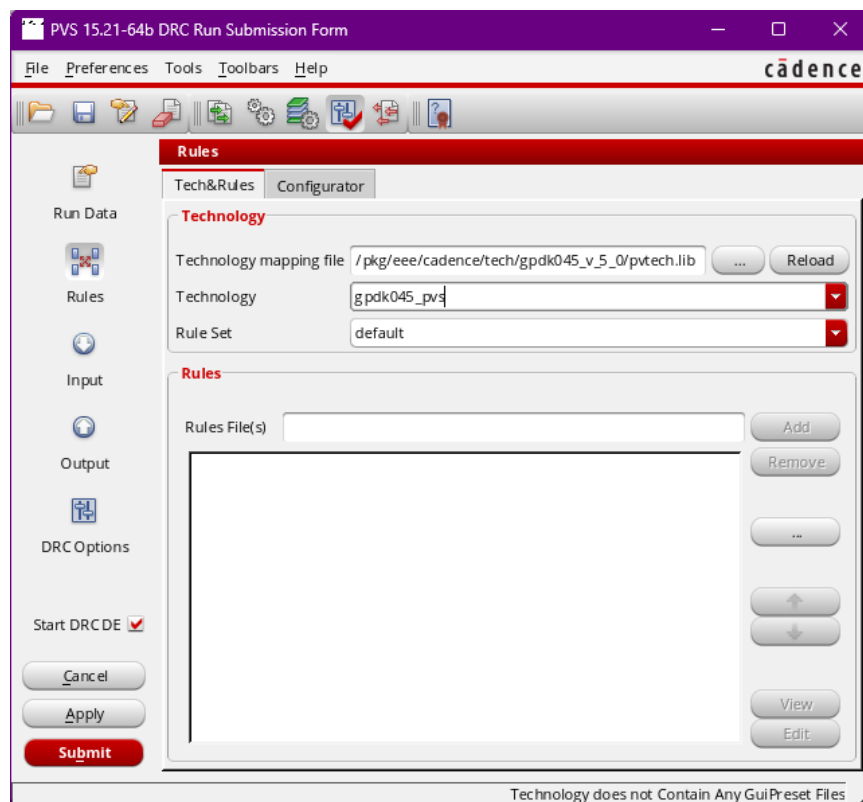


Figure 43: DRC Rules tab with the technology mapping file and gpdk045_pvs technology selected.

47. Click Submit to run the physical rules verification.

48. The first run typically reports one error: NIMP.X.W - “Nimp is not allowed over Pimp”, because the Nimp (yellow) and Pimp (red) layer rectangles are drawn too close together, resulting in an overlapping region at the center of the cell or over the opposing transistor’s body region.

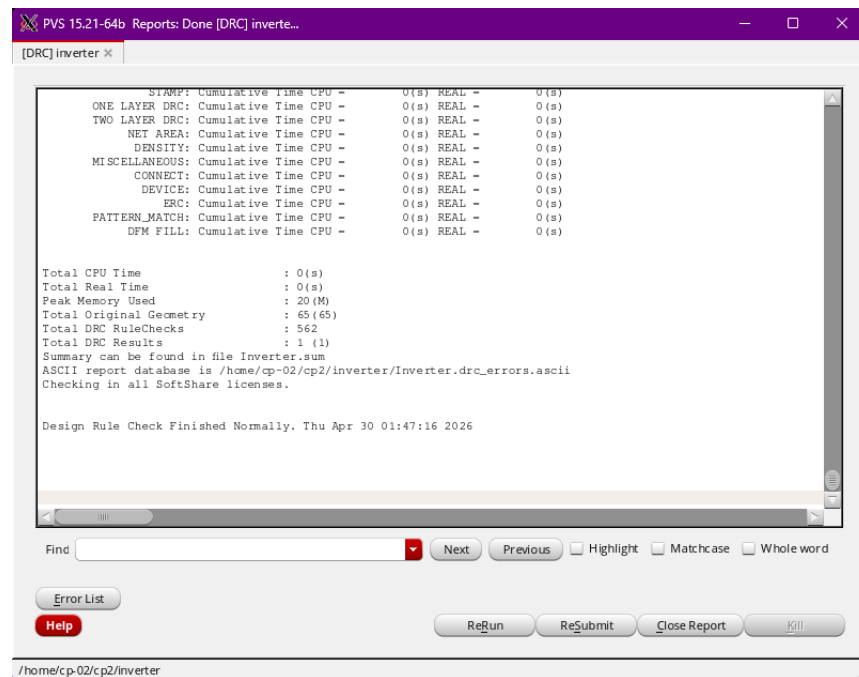


Figure 44: DRC report summary showing 1 rule violation (NIMP.X.W).

49. Open the DRC Debug Environment to identify and highlight the exact overlap region, and select View By Rule to inspect the NIMP.X.1 error.

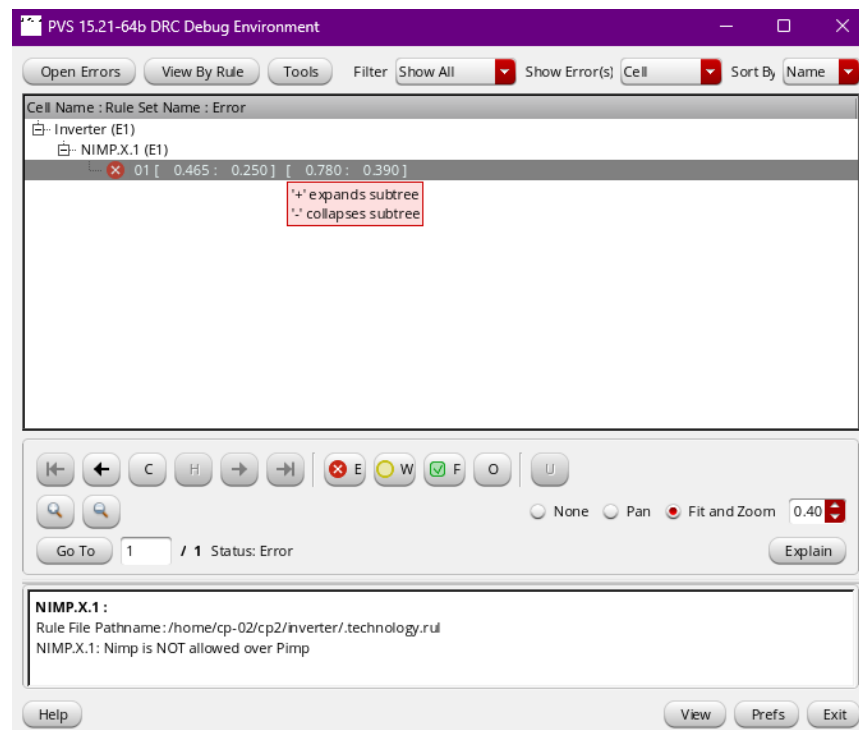


Figure 45: DRC Debug Environment showing the NIMP.X.1 error: “Nimp is NOT allowed over Pimp”.

50. Select the Nimp layer rectangle and use the Stretch tool (shortcut: S) to pull its top edge down, ensuring it is positioned entirely outside the Pimp area to avoid any overlap.

51. Rechecking - go to Plugins → PVS → Run DRC and submit the check again. Verify the updated layout returns 0 Errors, confirming the physical design is fully compliant with the technology rules.

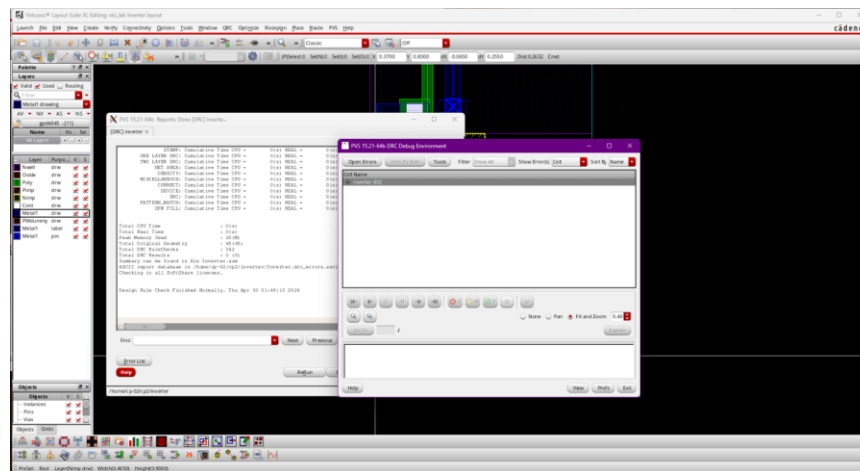


Figure 46: Re-run DRC report confirming 0 errors after the Nimp rectangle was stretched clear of the Pimp region.

Part 10 (Bonus): Layout Design of a Standard Cell - NAND Gate

The same standard-cell layout methodology used for the inverter also applies to multi-transistor gates such as a two-input NAND. The following steps illustrate how to align multiple transistors efficiently to minimize the overall cell area, using the NAND gate as an example.

How to Align Multiple Transistors (Area Minimization - NAND Gate)

1. In a NAND gate, the two PMOS transistors share a common drain node (OUT).
2. PMOS A (Gate A) initially has its Source on the left and its Drain on the right, and PMOS B (Gate B) initially has its Source on the left and its Drain on the right as well.

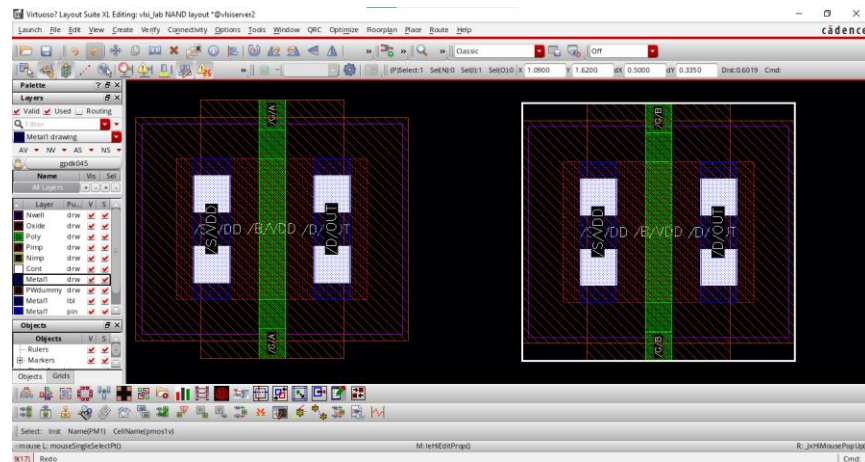


Figure 47: PMOS A and PMOS B placed side by side before alignment, each with Source on the left and Drain on the right.

3. To avoid routing long metal wires between the two separate drains, the second transistor must be flipped. Select the PMOS B transistor.
4. Go to the toolbox menu or use the Flip tool (horizontal mirror) to mirror the transistor.
5. After flipping, PMOS B now has its Drain on the left and its Source on the right.
6. Select the flipped PMOS B transistor.
7. Press A and click the flipped Drain edge of PMOS B, then align it directly with the Drain edge of PMOS A.
8. By overlapping these identical diffusion regions, the two separate drains merge into a single shared node. This eliminates the distance between the devices and significantly reduces the cell area.

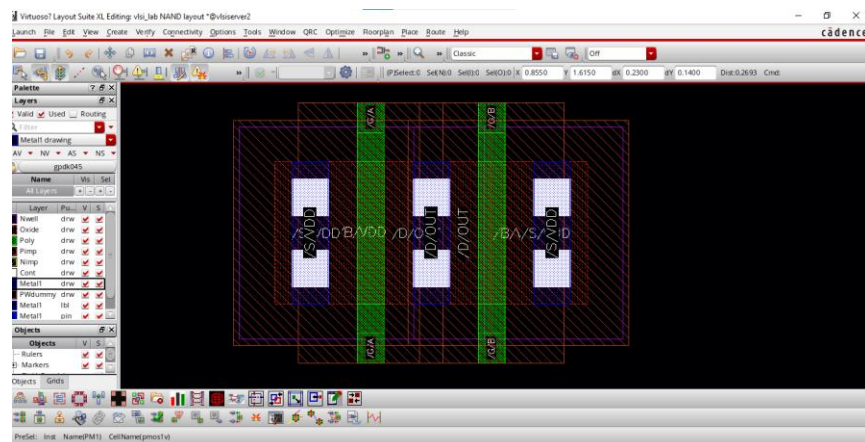


Figure 48: PMOS B flipped and aligned so its drain merges with the drain of PMOS A, sharing the OUT diffusion region.

9. In the NMOS pull-down network, the transistors are connected in series (the Source of NMOS B connects to the Drain of NMOS A).
10. Position the two NMOS transistors adjacent to each other. Press A to overlap the Source diffusion of NMOS B directly with the Drain diffusion of NMOS A.

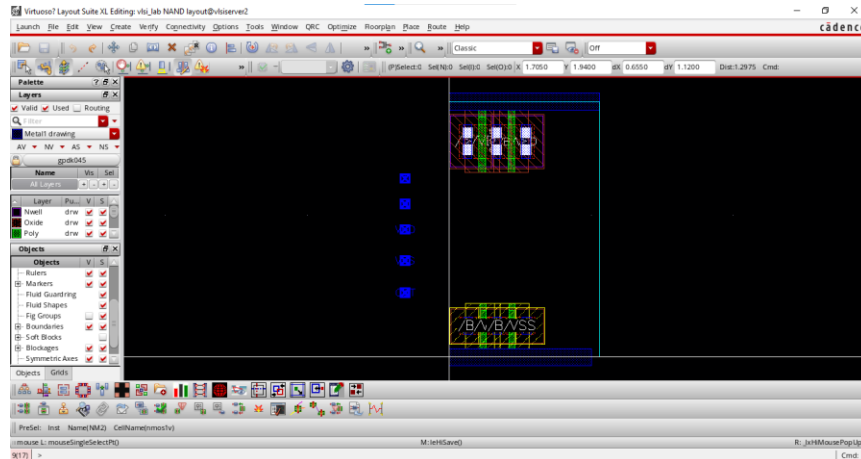


Figure 49: NMOS transistors positioned in series, with the shared source/drain diffusion merged.

11. Next, press A to use the Align tool to vertically align Gate A of the PMOS directly over Gate A of the NMOS.
12. Select both the right-aligned PMOS and NMOS transistors, press Q to open the properties menu, and change the Body Type parameter to detached. Click right-tap.

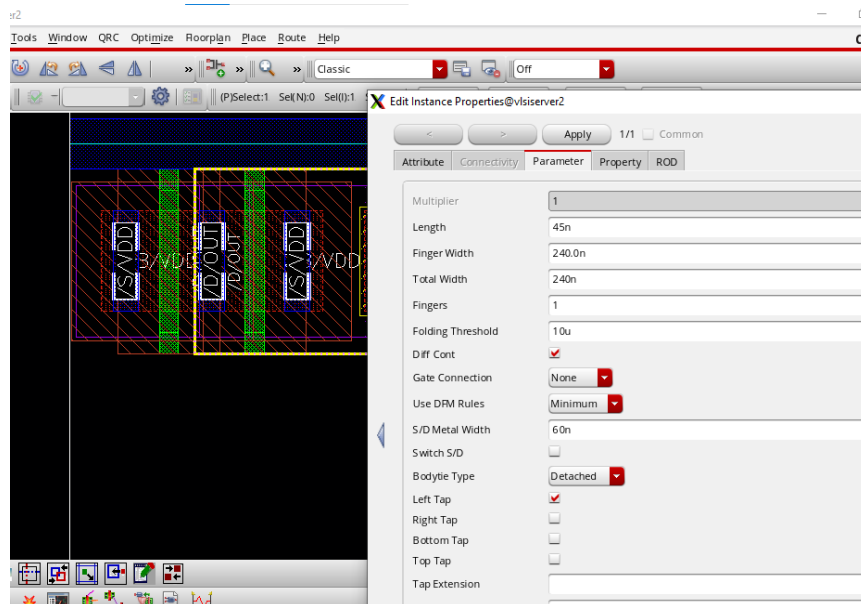


Figure 50: Edit Instance Properties for the right-hand PMOS, with Body Type set to Detached.

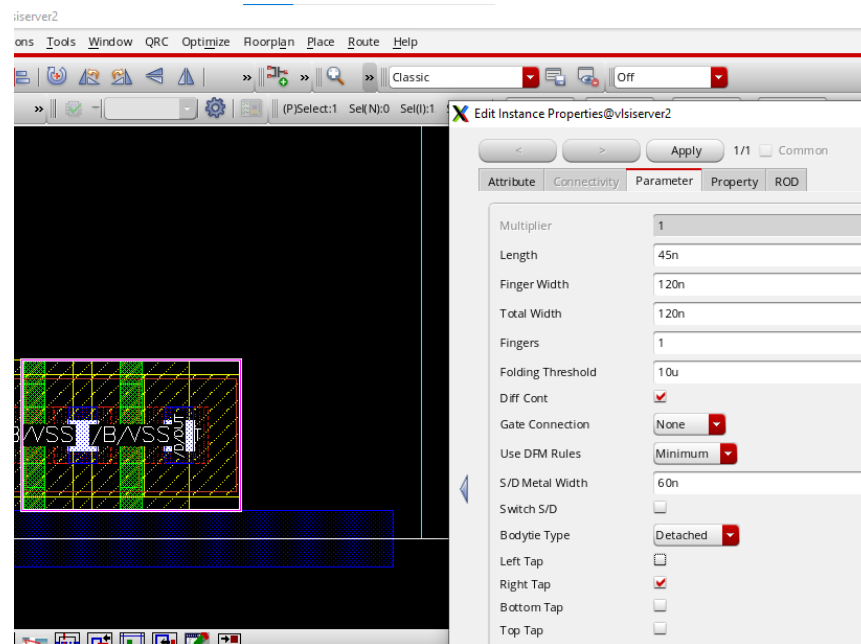


Figure 51: Edit Instance Properties for the right-hand NMOS, with Body Type set to Detached.

Layout Design of a Standard Cell - NAND Gate

Now, following the same procedure used for the inverter, here is the physical layout for the NAND gate, followed by its corresponding LVS and DRC verification steps.

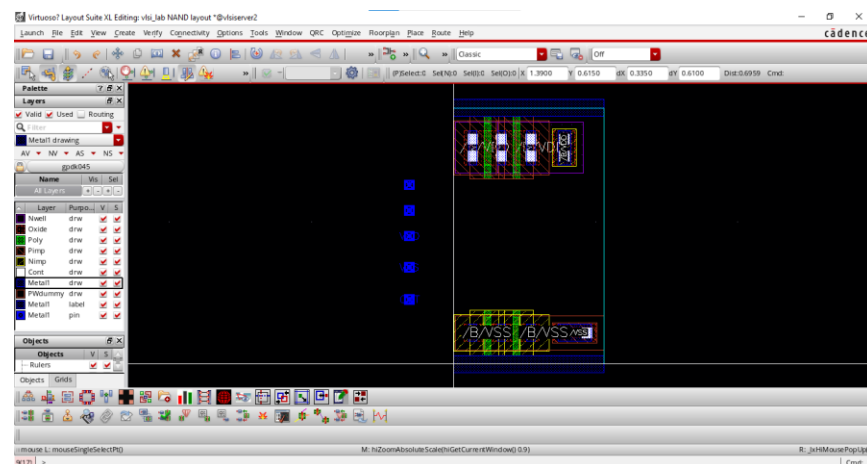


Figure 52: NAND gate layout with VDD/VSS tie routing complete for all transistors.

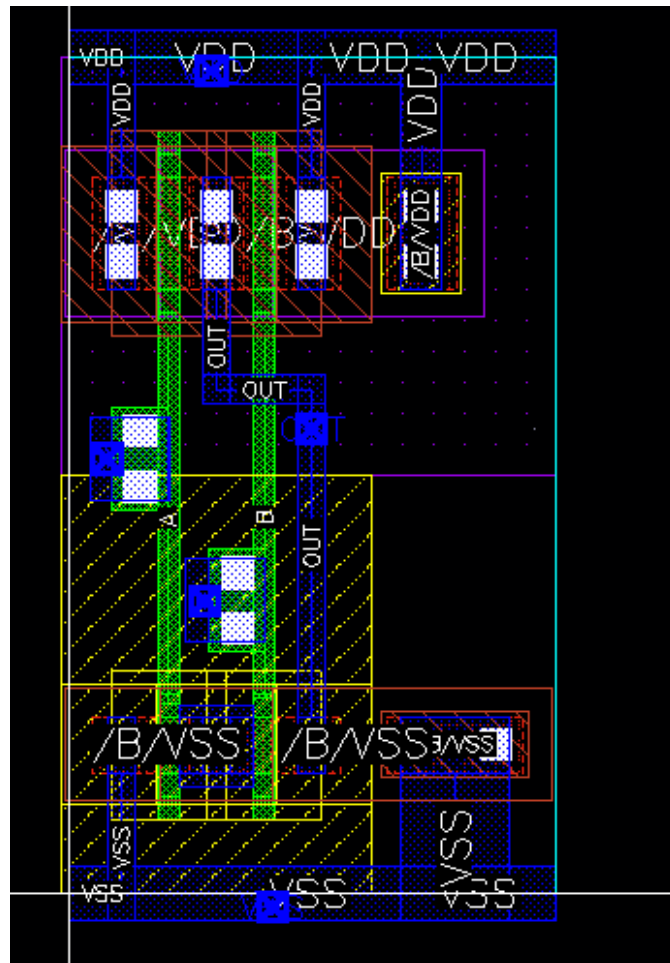


Figure 53: Complete NAND gate standard-cell layout, with the N-Well, implant layers, gate routing and power rails in place.

DRC Check

Follow the same steps as for the inverter to run DRC on the NAND gate layout.

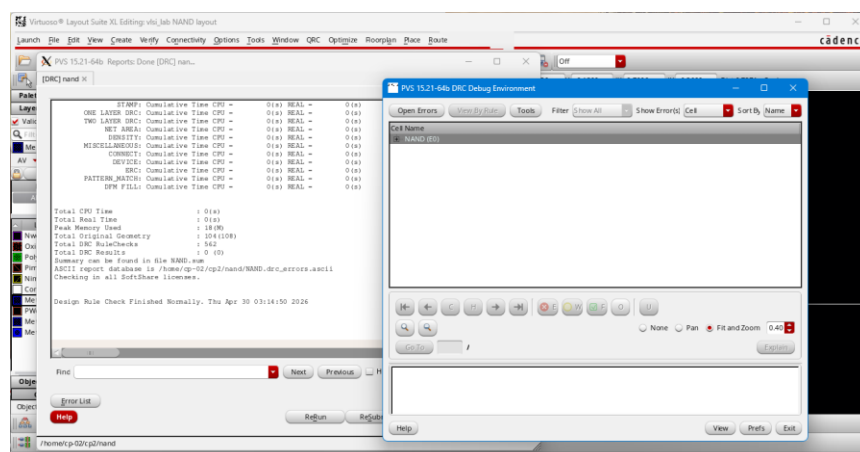


Figure 54: NAND gate DRC report confirming 0 errors.

LVS Check

Follow the same steps as for the inverter to run LVS on the NAND gate layout.

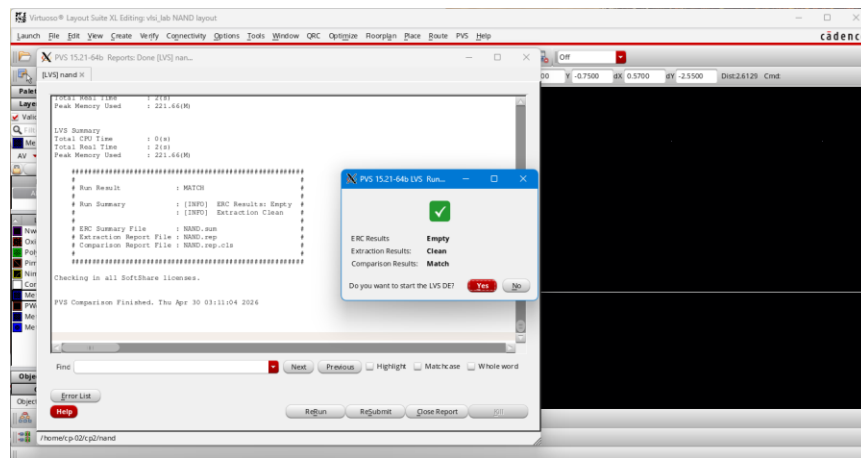


Figure 55: NAND gate LVS run result confirming Comparison Results: Match.

Homework

1. Draw the physical layout of the Inverter, NAND, NOR, and XOR logic circuits, following the exact standard-cell design steps outlined above.

Lab Report / Submission Checklist

1. A screenshot of your completed inverter layout, with the PR boundary, VDD/VSS rails, and all pins labelled.
2. A screenshot of your generated inverter symbol-to-layout correspondence (LVS Match result).
3. A screenshot of the final DRC report showing 0 errors for the inverter layout.
4. A short discussion (150-200 words) explaining the purpose of the N-Well, body-tap detachment, and the DRC/LVS verification flow in standard-cell design.
5. (Optional/Bonus) Screenshots of your NAND gate layout, and its DRC and LVS results.

Credits

EEE, UIU

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